HIREGEN AI

AI RECRUITMENT & CANDIDATE ASSESSMENT PLATFORM

Platform Enhancement Research

AI hiring trends, architecture notes and feature prioritisation for the next phase of HireGen AI.

Document HireGen-RES-001

Version v1.0

Author Irham Saleha

Prepared for HireGen AI project documentation

1. PURPOSE

HireGen AI shipped its first full-stack release with a generator–critic assessment engine, an AI requirement parser, and a production service layer around both. This document captures the research behind what to build next: where AI hiring tools are trending, what the current architecture makes easy or hard, and how the resulting feature candidates were prioritised.

2. AI HIRING TRENDS

Screening is moving from keyword matching to rubric-based scoring.

- Recruiters increasingly expect a visible, explainable score rather than a black-box rank — matching HireGen's existing AI Fit Score and explainability panel.
- Rubrics generated from the job description itself (rather than a generic template) are seen as materially more trustworthy by hiring managers.

Generated assessment content needs a human-in-the-loop step.

- Fully automated MCQ and behavioural-question generation is now common, but unreviewed output damages trust after a small number of visibly wrong questions.
- A generator–critic pattern with over-generation and filtering (as HireGen already runs at a 2× buffer) is becoming the expected minimum bar, not a differentiator on its own.

Provider risk is now a product concern, not just an infra concern.

- Single-provider LLM dependence is treated as a reliability risk by technical buyers; model rotation and fallback handling are increasingly asked about directly in vendor evaluations.
- Throttling and graceful degradation (returning a smaller, still-useful result rather than failing) are now baseline expectations under load.

Authenticity and integrity checks are rising in priority.

- AI-assisted candidate answers have made plagiarism and authenticity flags a frequently requested feature, not an edge case.
- Buyers want flags to be inspectable (evidence attached) rather than a bare pass/fail signal.

3. ARCHITECTURE NOTES

The current build spans 10 schemas, 8 controllers and 9 service modules, with the AI service layer already isolated behind model rotation, throttling and fallback handling. That isolation is the reason the items below are additive rather than structural.

Area	Observation
AI service layer	Already decoupled from controllers via rotation/throttling/fallback — new models or providers can be added without touching calling code.
Requirement parser	Runs once per job description and persists structured output — a natural point to add versioning if criteria are edited after posting.

Assessment pipeline	Generator–critic buffer-and-filter is synchronous per batch — a queue would let large batches run without blocking the request.
RBAC	JWT-based roles are enforced per endpoint — supports adding new scoped roles (e.g. read-only interviewer) without a schema change.

4. FEATURE PRIORITISATION

Candidates were scored on impact to recruiter trust and time-to-hire against implementation effort, given the architecture notes above.

Feature	Impact	Effort	Priority
Explainability evidence on every AI flag	High	Low	P0
Assessment generation queue for large batches	Medium	Medium	P1
Requirement-parser versioning on edit	Medium	Low	P1
Additional model provider for fallback	High	Medium	P1
Read-only interviewer role	Low	Low	P2
Candidate-facing score appeal workflow	Medium	High	P2

5. RECOMMENDATIONS

- Attach supporting evidence to every risk flag and authenticity badge before adding new detection signals — trust in existing flags compounds faster than new flag types.
- Move large-batch assessment generation onto a queue so throughput scales independently of request latency.
- Add a second model provider behind the existing rotation layer as the next reliability investment, since the abstraction already exists.
- Defer the score-appeal workflow until explainability evidence ships — an appeal without visible evidence just moves the trust problem downstream.